

Detection and Precision Tracking System

Background and System Overview

You are given the following video in which you can see a person walking in a natural outdoor environment: <https://youtube.com/shorts/wXDXFysTJlk?si=ykHQJNwg2my1s8cp>

In this scenario, the target may be partially or fully occluded for periods of time, may exhibit appearance changes, and may be difficult to distinguish from the background. The system therefore needs to rely on both spatial and temporal reasoning.

System Goals

- **Real-Time Detection:** Immediate identification of the target under occlusion and background clutter.
- **Real-Time Tracking:** Maintain continuous tracking and preserve target identity despite occlusions and visibility changes
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Deliverables

1. **System Design:** Provide a high-level architecture of the system. Clearly define all functional blocks and their inputs/outputs. Include at least one Deep Learning-based block. For each block, please define the inputs and outputs, algorithms, assumptions, trade-offs, and potential risks/limitations.
2. **Deep Learning Analysis:** For the block you've decided to implement using deep learning, please provide a literature review (2-3 relevant approaches). You need to explain model selection and justify why it fits this problem, pros/cons, limitations and any other relevant information for the decision.
3. **Success Criteria:** Define clear measurable metrics for each block in the system, including end-to-end metrics.
4. **False Analysis:** provide a structured analysis of failure modes in the system, their root causes, and their impact.

5. **Improvement Suggestions:** Propose improvements and explain the reasoning behind them.

Submission Guidelines

- Format: PDF or slides (max **6 pages or 15 slides**)
- Attach running code or a GitHub link
- Provide a video demonstrating the detection and tracking results